ARCHITECTURAL JUSTIFICATION

System: Healthcare Bed Allocation / Emergency Room Bed Management System
Selected Architecture: Layered Architecture

Architecture Selection:
We chose **Layered Architecture** for the Healthcare Bed Allocation / Emergency Room Bed Management System. The system is organized into Presentation, Business, and Data layers, with each layer handling a specific responsibility.

Reason 1 – Clear Separation of Responsibilities:
Layered Architecture separates the user interface from the business logic and data storage. The Presentation layer handles the dashboard used by the Triage Nurse and Hospital Administrator, while the Business layer handles triage, bed allocation, alerts, and transfer management. The Data layer stores patient, bed, allocation, and related information. This makes the system easier to understand, maintain, and modify.

Reason 2 – Suitable for the System:
The Healthcare Bed Allocation System is an integrated hospital application where the different functions need to work together. Layered Architecture provides a simple and structured design for these functions without introducing the additional operational complexity of a Microservices Architecture.

Security Advantage:
The architecture improves security by keeping sensitive patient and allocation data behind the Business layer. The user interface does not directly access the database, allowing access to patient and bed information to be controlled through the application logic.

Performance Benefit:
Layered Architecture keeps the main system operations within a structured application flow and avoids unnecessary network communication between multiple independent services. This supports quick bed-status updates and helps the system meet the requirement for real-time updates.

Conclusion:
Therefore, Layered Architecture is suitable for the Healthcare Bed Allocation / Emergency Room Bed Management System because it provides clear separation of responsibilities, supports secure data access, and enables efficient handling of critical bed allocation operations.